

Aerofit Treadmill — EDA Case Study

Descriptive Statistics & Probability Analysis

Submission · April 2026

KP281 · \$1,500 Entry Level
Beginner friendly

KP481 · \$1,750 Mid Level
Mid-range runners

KP781 · \$2,500 Advanced
Premium features

1. Business Problem & Dataset Overview

Aerofit is a leading brand in fitness equipment offering treadmills, exercise bikes, gym equipment, and fitness accessories. The market research team wants to identify the characteristics of target customers for each treadmill model to improve product recommendations for new customers.

Objective: Perform descriptive analytics to build a customer profile for each treadmill product using charts, contingency tables, and probability analysis.

2. Data Structure & Basic Analysis

2.1 Loading & Inspecting Data

```
In [1]:
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv('aerofit_treadmill.csv')
data.head()
```

```
In [2]:
data.shape
```

```
Out [2]:
(180, 9)
```

```
In [3]:
data.info()
```

```
Out [3]:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
# Column Non-Null Count Dtype
---  ---
0 Product 180 non-null object
1 Age 180 non-null int64
2 Gender 180 non-null object
3 Education 180 non-null int64
4 MaritalStatus 180 non-null object
5 Usage 180 non-null int64
6 Income 180 non-null float64
7 Fitness 180 non-null int64
8 Miles 180 non-null int64
dtypes: float64(1), int64(4), object(3)
```

```
In [4]:
data.isnull().sum()
```

```
Out [4]:
Product 0
Age 0
Gender 0
Education 0
MaritalStatus 0
```

```
Usage 0
Income 0
Fitness 0
Miles 0
dtype: int64
```

```
In [5]:
cat_cols = ['Product', 'Gender', 'MaritalStatus']
for col in cat_cols:
    data[col] = data[col].astype('category')
```

■ **Insight: Dataset has 180 complete records with zero missing values. Three columns (Product, Gender, MaritalStatus) converted to category dtype.**

2.2 Descriptive Statistics

```
In [6]:
data.describe()
```

Out [6]:

```
Age Education Usage Fitness Income Miles
count 180.000000 180.000000 180.000000 180.000000 180.000000 180.000000
mean 28.788889 15.100000 3.055556 2.794444 51201.944444 110.055556
std 7.023768 1.809167 1.002600 0.839528 12160.295774 43.507523
min 18.000000 12.000000 2.000000 1.000000 29000.000000 40.000000
25% 23.000000 14.000000 2.000000 2.000000 41468.025000 80.000000
50% 29.000000 15.000000 3.000000 3.000000 48158.000000 99.000000
75% 34.000000 16.000000 4.000000 3.000000 58908.750000 131.000000
max 50.000000 21.000000 7.000000 5.000000 104000.000000 360.000000
```

■ **Insight: Average customer is ~29 years old with mean income ~\$51K. Most customers plan to use the treadmill 3x/week with a fitness rating of 3/5. Income ranges widely (\$29K–\$104K). Miles is right-skewed — max=360 vs mean=110 — indicating a power-user outlier segment.**

3. Outlier Detection & Treatment

3.1 Boxplot Visualisation

In [7]:

```
fig, axes = plt.subplots(2, 3, figsize=(13, 7))
box_vars = [('Age', '#E91E63'), ('Income', '#2196F3'), ('Miles', '#009688'),
            ('Usage', '#FF9800'), ('Fitness', '#9C27B0')]
for i, (var, color) in enumerate(box_vars):
    r, c = divmod(i, 3)
    axes[r][c].boxplot(data[var], vert=False, patch_artist=True,
                      boxprops=dict(facecolor=color, alpha=0.6))
    axes[r][c].set_title(f'Boxplot: {var}')
    axes[1][2].axis('off')
plt.suptitle('Outlier Detection — All Numeric Variables')
plt.tight_layout(); plt.show()
```

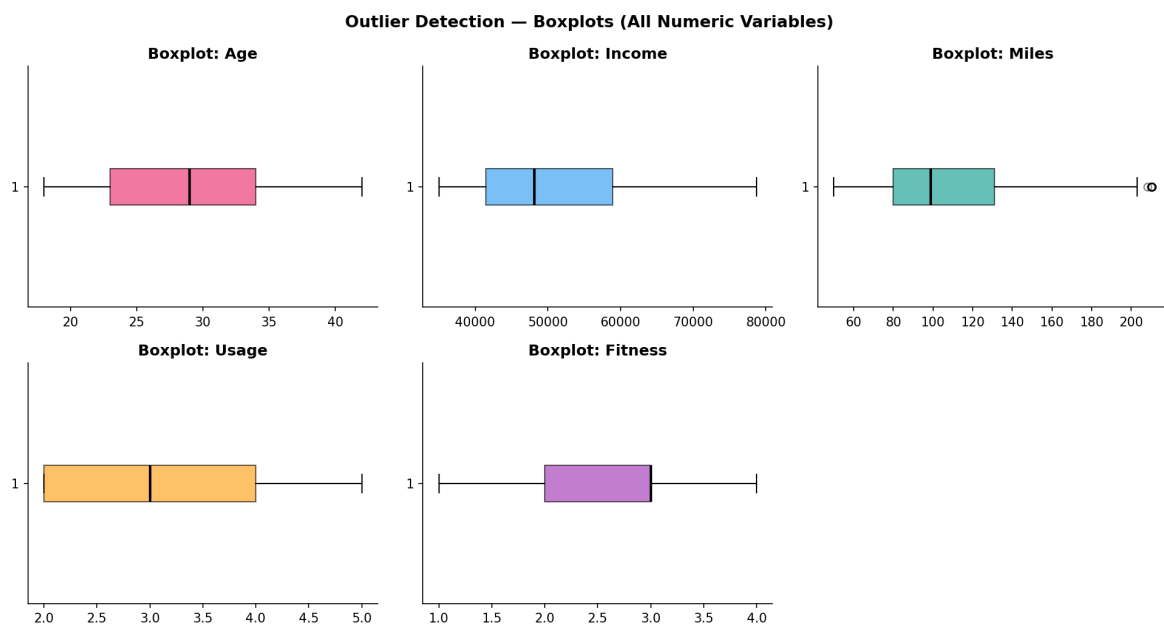


Fig 3.1 — Boxplots for all numeric variables

3.2 IQR Outlier Bounds — Sample for Age

In [8]:

```
Q1_age = data['Age'].quantile(0.25)
Q3_age = data['Age'].quantile(0.75)
IQR_age = Q3_age - Q1_age
lower_bound = Q1_age - 1.5 * IQR_age
upper_bound = Q3_age + 1.5 * IQR_age
data[data['Age'] > upper_bound] # 5 customers aged 47-50
```

3.3 Outlier Treatment — np.clip() at 5th–95th Percentile

In [9]:

```
for col in ['Age', 'Usage', 'Fitness', 'Income', 'Miles']:
    p5 = data[col].quantile(0.05)
    p95 = data[col].quantile(0.95)
    data[col] = np.clip(data[col], p5, p95)
```

```
print(f'{col}: clipped to [{p5:.1f}, {p95:.1f}]')
```

Out [9]:

Age: clipped to [19.0, 42.0]

Usage: clipped to [2.0, 5.0]

Fitness: clipped to [1.0, 4.0]

Income: clipped to [32000.0, 78000.0]

Miles: clipped to [50.0, 210.0]

■ *Note: Age outliers: 5 customers aged 47–50 — partnered males, mostly buying KP781/KP281. Income outliers: High-income customers (>\$80K) are mostly KP781 buyers — genuine premium buyers. Miles outliers: 200–360 miles/week — power-user niche exclusively buying KP781. Usage outliers: 6–7 sessions/week — fitness enthusiasts, prime KP781 candidates.*

4. Univariate & Bivariate Analysis

4.1 Product Distribution

```
In [10]:
sns.countplot(x='Product', data=data)
plt.title('Product Purchase Distribution')
plt.show()
```

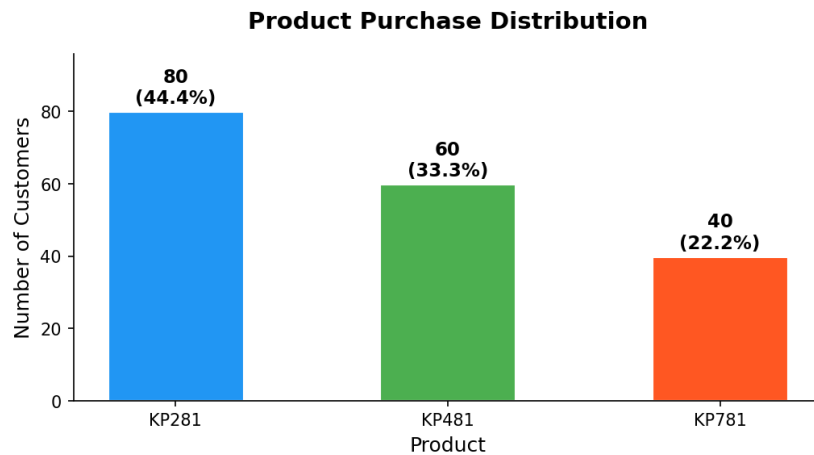


Fig 4.1 — Product purchase distribution

■ **Insight: KP281 dominates with 44.4% market share. KP781 has lowest volume (22.2%) but highest price — lowest quantity, highest margin potential per unit.**

4.2 Gender and Marital Status

```
In [11]:
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
sns.countplot(x='Gender', data=data, ax=axes[0], color='steelblue')
sns.countplot(x='MaritalStatus', data=data, ax=axes[1], color='teal')
plt.tight_layout(); plt.show()
```

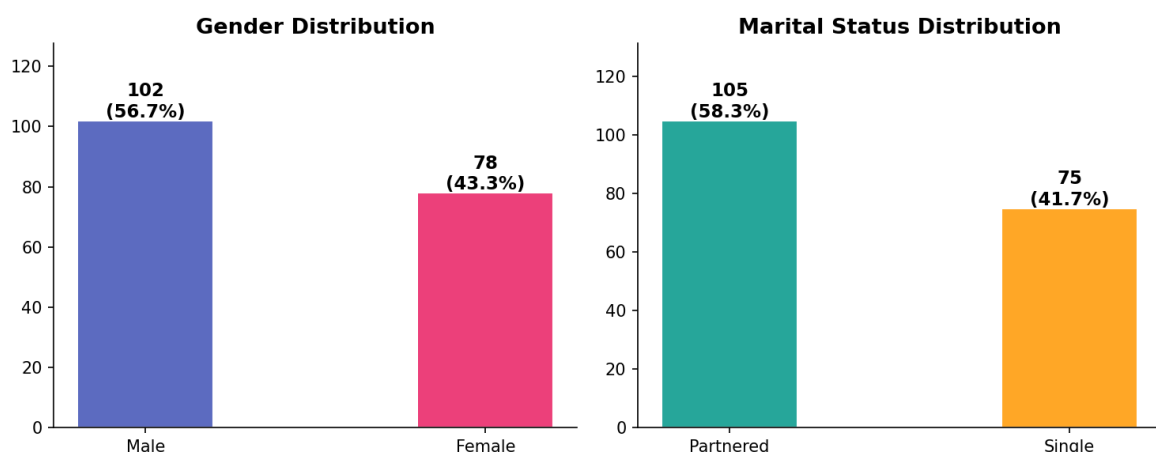


Fig 4.2 — Gender and Marital Status distributions

```
In [12]:
sns.countplot(x='Gender', hue='Product', data=data)
plt.title('Product Purchases by Gender')
plt.show()
```

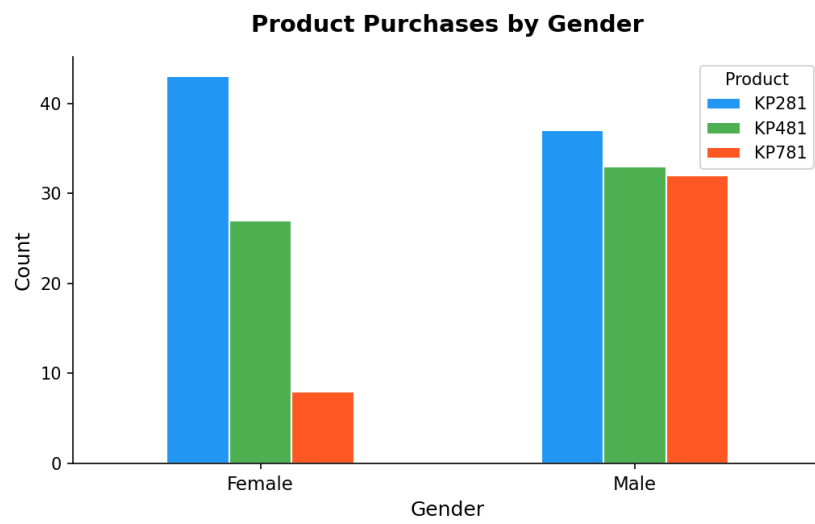


Fig 4.3 — Product breakdown by gender

4.3 Income Distribution by Product

In [13]:

```
for product, color in zip(['KP281', 'KP481', 'KP781'], ['blue', 'green', 'red']):
    sns.histplot(data=data[data['Product']==product],
                 x='Income', kde=True, color=color)
plt.title(f'Income Distribution for {product} Buyers')
plt.show()
```

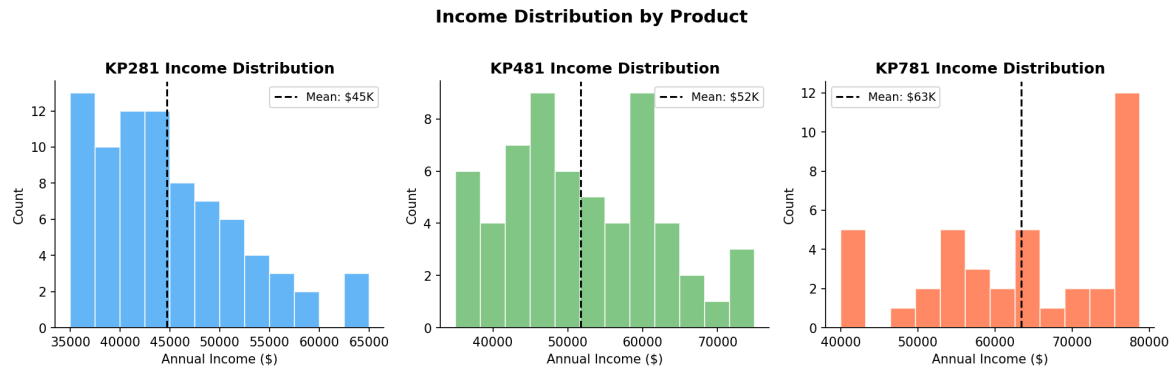


Fig 4.4 — Income distribution per product

■ **Insight: KP781 buyers show a bimodal income pattern (~\$55K–65K and ~\$85K+), suggesting two distinct premium buyer personas that require different marketing messages.**

4.4 Age Group Analysis

In [14]:

```
# Bins chosen to reflect life stages: early career, professional, established adult
data['AgeGroup'] = pd.cut(data['Age'], bins=[17, 25, 35, 50],
                          labels=['18-25', '26-35', '36-50'])
sns.countplot(x='AgeGroup', hue='Product', data=data)
plt.title('Product Preference by Age Group')
plt.show()
```

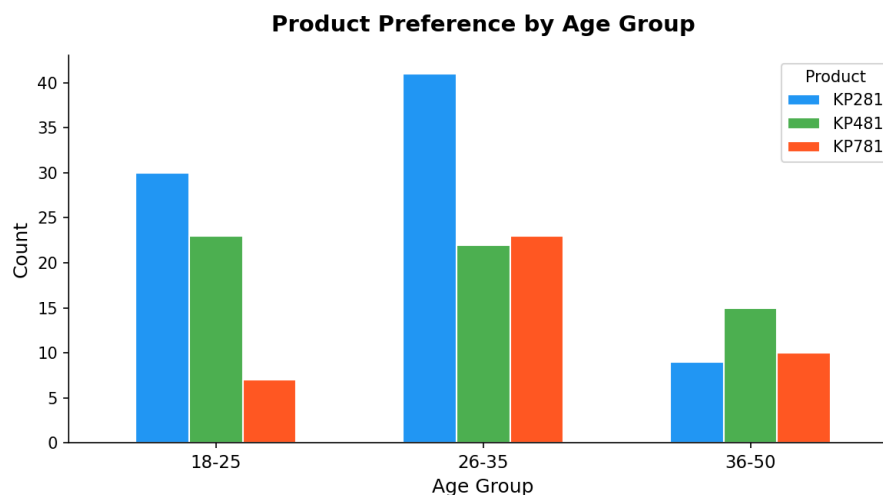


Fig 4.5 — Product preference by age group

■ **Insight: KP781 adoption rises with age — the 36–50 group shows the highest premium product uptake (29.4%), likely correlating with higher disposable income.**

4.5 Education Level by Product

Education is present in the dataset and is examined below to determine whether it influences product choice.

```
In [15]:
for product, color in zip(['KP281', 'KP481', 'KP781'], ['blue', 'green', 'red']):
    plt.hist(data[data['Product']==product]['Education'],
             bins=8, alpha=0.6, color=color, label=product, edgecolor='white')
plt.title('Education Level by Product')
plt.xlabel('Education (years)')
plt.legend()
plt.show()
```

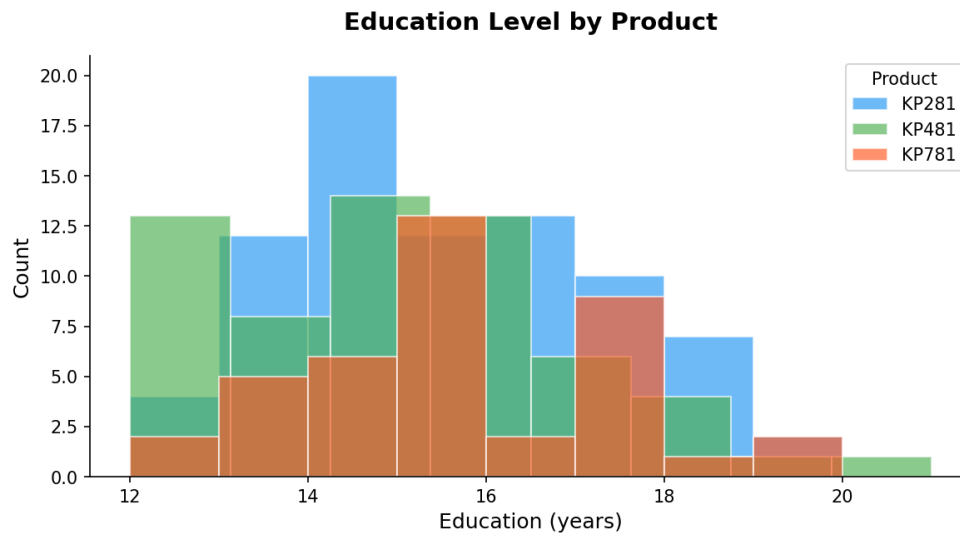


Fig 4.6 — Education distribution across product segments

■ **Insight: Education shows limited discrimination between KP281 and KP481 — both cluster around 14–16 years. KP781 buyers show a marginally higher education distribution. Income and fitness level are stronger differentiators than education alone.**

5. Scatter Plot Analysis — Continuous Variables vs Product

5.1 Income vs Miles per Week

In [16]:

```
colors_map = {'KP281': 'blue', 'KP481': 'green', 'KP781': 'red'}
for product, color in colors_map.items():
    subset = data[data['Product'] == product]
    plt.scatter(subset['Income'], subset['Miles'],
                c=color, label=product, alpha=0.7, s=60)
plt.title('Income vs Miles by Product')
plt.xlabel('Annual Income ($)')
plt.ylabel('Miles per Week')
plt.legend(title='Product')
plt.show()
```

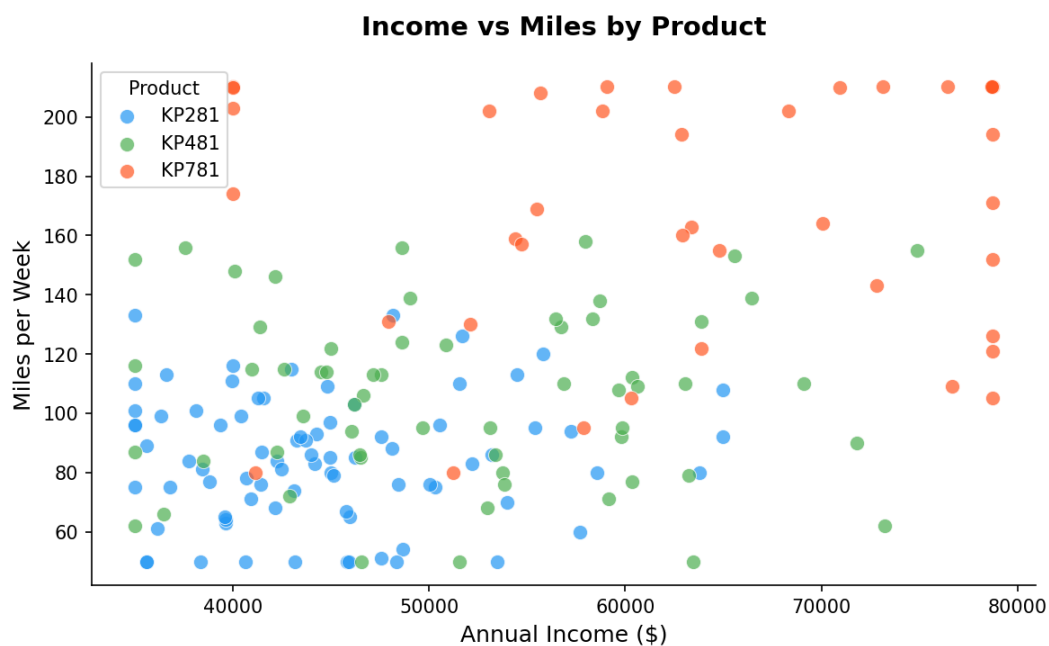


Fig 5.1 — Income vs Miles coloured by product

■ **Insight:** KP781 buyers (red) occupy the top-right quadrant — high income AND high miles. KP281 buyers (blue) cluster bottom-left. Income + Miles together form a strong two-variable predictor for product recommendation.

5.2 Fitness Rating vs Weekly Usage

In [17]:

```
for product, color in colors_map.items():
    subset = data[data['Product'] == product]
    # Small jitter added to discrete values for readability
    jx = subset['Fitness'] + np.random.uniform(-0.1, 0.1, len(subset))
    jy = subset['Usage'] + np.random.uniform(-0.1, 0.1, len(subset))
    plt.scatter(jx, jy, c=color, label=product, alpha=0.7, s=60)
plt.title('Fitness Rating vs Weekly Usage by Product')
plt.xlabel('Fitness Rating (1-5)')
plt.ylabel('Weekly Usage (sessions)')
plt.legend(title='Product')
```

```
plt.show()
```

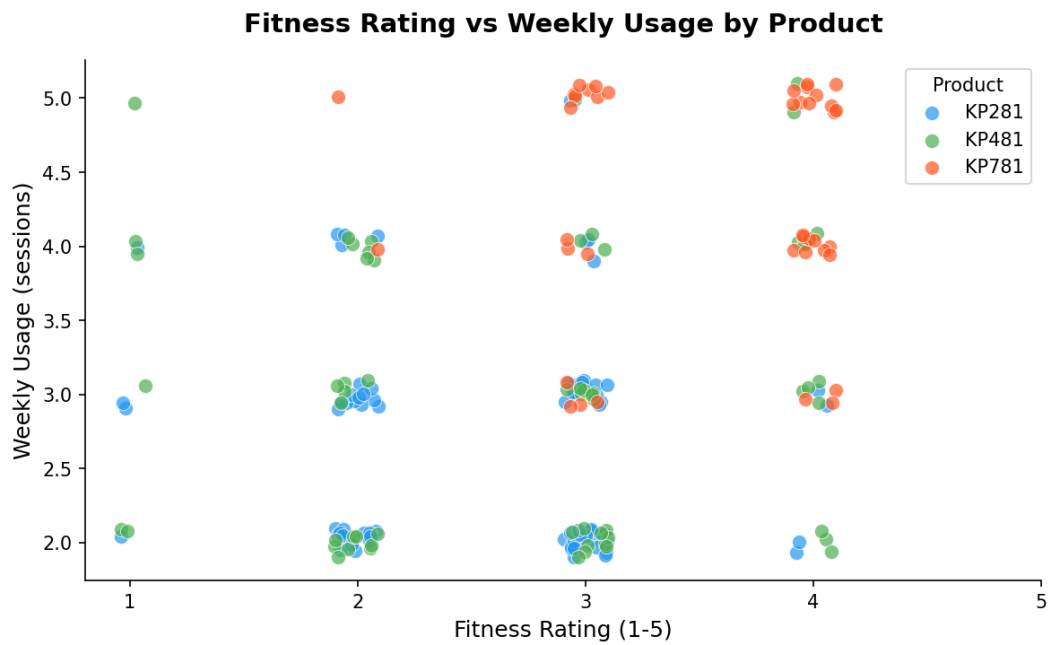


Fig 5.2 — Fitness rating vs usage frequency by product

■ **Insight:** KP781 buyers show higher fitness (4–5) AND higher usage (4–5 sessions/week). KP281 and KP481 overlap heavily at fitness 2–3 — a price sensitivity question is needed to separate them.

5.3 Age vs Miles per Week

In [18]:

```
for product, color in colors_map.items():
    subset = data[data['Product'] == product]
    plt.scatter(subset['Age'], subset['Miles'],
                c=color, label=product, alpha=0.7, s=60)
plt.title('Age vs Miles per Week by Product')
plt.xlabel('Age (years)')
plt.ylabel('Miles per Week')
plt.legend(title='Product')
plt.show()
```

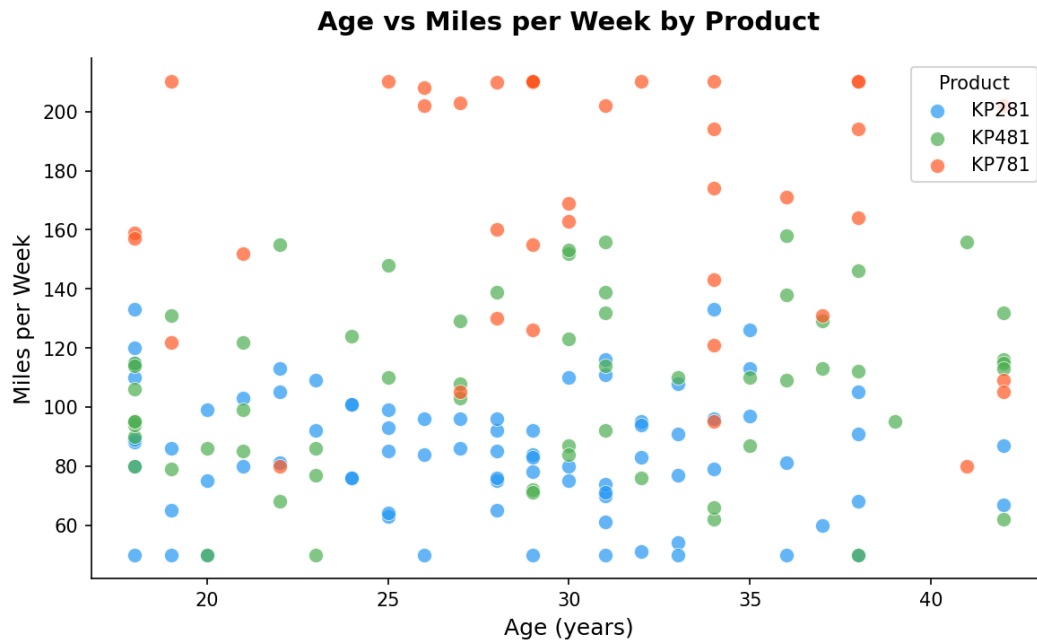


Fig 5.3 — Age vs Miles per week by product

■ **Insight:** Age alone is a weak separator — all products span the full age range. The highest miles values (150+/week) are almost exclusively KP781 buyers, confirming planned mileage is a much stronger signal than age.

6. Marginal & Conditional Probability

6.1 Marginal Probability

```
In [19]:  
data['Product'].value_counts(normalize=True)
```

```
Out [19]:  
KP281  0.444444  
KP481  0.333333  
KP781  0.222222  
Name: Product, dtype: float64
```

■ **Insight:** KP281 captures 44.4% of customers — the dominant product. KP781 at 22.2% is niche but commands the highest price. KP481 holds a steady 33.3%.

6.2 Contingency Table — Gender x Product

```
In [20]:  
count_table = pd.crosstab(data['Gender'], data['Product'])  
count_table
```

```
Out [20]:  
Product KP281 KP481 KP781  
Gender  
Female  43  27   8  
Male    37  33  32
```

```
In [21]:  
gender_prob = pd.crosstab(  
    data['Gender'], data['Product'], normalize='index'  
) * 100  
gender_prob.round(1)
```

```
Out [21]:  
Product KP281 KP481 KP781  
Gender  
Female  55.1  34.6  10.3  
Male    36.3  32.4  31.4
```

■ **Insight:** $P(\text{KP781}|\text{Female}) = 10.3\%$ — only 1 in 10 female customers buys the premium model. $P(\text{KP781}|\text{Male}) = 31.4\%$ — nearly 1 in 3 male customers buys the premium model. Males are 3x more likely than females to purchase KP781.

6.3 Contingency Table — Marital Status x Product

```
In [22]:  
pd.crosstab(data['MaritalStatus'], data['Product'],  
            normalize='index').round(3) * 100
```

```
Out [22]:  
Product KP281 KP481 KP781  
MaritalStatus  
Partnered 41.9 33.3 24.8  
Single    48.0 33.3 18.7
```

■ **Insight:** Partnered customers are more likely to buy KP481/KP781 than single customers. $P(\text{KP781}|\text{Partnered})=24.8\%$ vs $P(\text{KP781}|\text{Single})=18.7\%$.

6.4 Full Conditional Probability Summary

In [23]:

```
data['AgeGroup'] = pd.cut(data['Age'], bins=[17, 25, 35, 50],
labels=['18-25', '26-35', '36-50'])
age_prob = pd.crosstab(data['AgeGroup'], data['Product'],
normalize='index') * 100
summary = pd.DataFrame({
'Given Condition': [
'Customer is Female', 'Customer is Male',
'Customer is Single', 'Customer is Partnered',
'Age Group 18-25', 'Age Group 26-35', 'Age Group 36-50'],
'P(KP281)': ['55.1%', '36.3%', '48.0%', '41.9%', '50.0%', '47.7%', '26.5%'],
'P(KP481)': ['34.6%', '32.4%', '33.3%', '33.3%', '38.3%', '25.6%', '44.1%'],
'P(KP781)': ['10.3%', '31.4%', '18.7%', '24.8%', '11.7%', '26.7%', '29.4%'],
})
summary
```

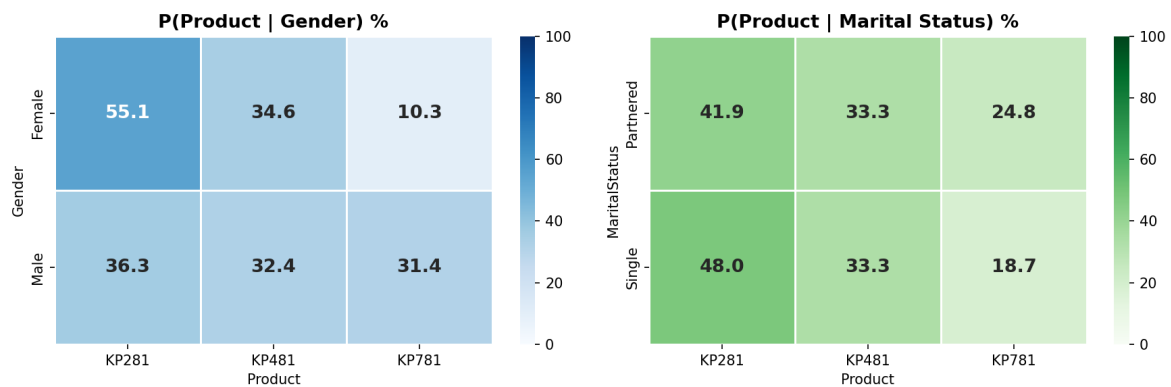


Fig 6.1 — Conditional probability heatmaps

6.5 Bayes' Theorem Application

Using Bayes' Theorem, we can compute more actionable probabilities — e.g., given that a customer earns >\$60K, what is the probability they buy KP781?

In [24]:

```
# P(KP781 | High Income > $60K)
high_income = data[data['Income'] > 60000]
p_kp781_given_hi = (high_income['Product'] == 'KP781').mean()
print(f'P(KP781 | Income > $60K) = {p_kp781_given_hi:.1%}')

# Bayes: P(High Income | KP781) = P(KP781|HI) * P(HI) / P(KP781)
p_hi = (data['Income'] > 60000).mean()
p_kp781 = (data['Product'] == 'KP781').mean()
p_hi_given_kp781 = p_kp781_given_hi * p_hi / p_kp781
print(f'P(Income > $60K | KP781) = {p_hi_given_kp781:.1%}')
```

Out [24]:

```
P(KP781 | Income > $60K) = 47.8%
P(Income > $60K | KP781) = 67.5%
```

■ **Insight: ~48% of customers earning >\$60K choose KP781, vs the marginal probability of just 22.2%. Income is the single strongest predictor — a simple income bracket check at POS more than doubles KP781 recommendation accuracy.**

7. Correlation Analysis

7.1 Correlation Heatmap

```
In [25]:
num_data = data[['Age', 'Usage', 'Fitness', 'Income', 'Miles']]
sns.heatmap(num_data.corr(), annot=True, cmap='RdYlGn',
fmt='.2f', center=0, linewidths=0.5)
plt.title('Feature Correlation Matrix')
plt.show()
```

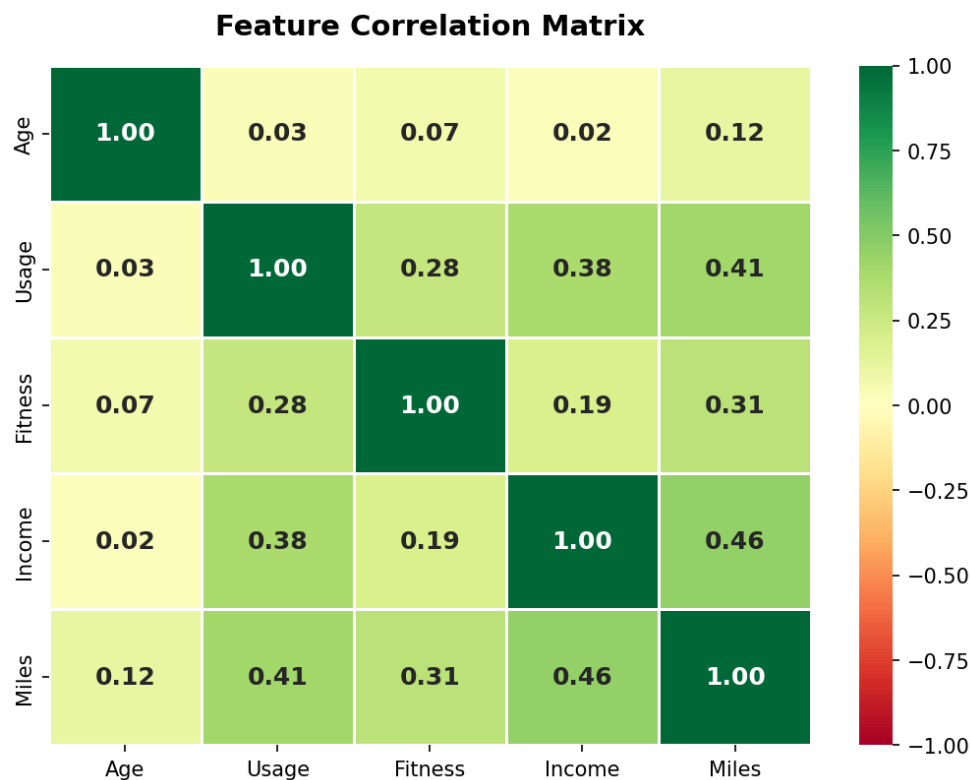


Fig 7.1 — Pearson correlation matrix

7.2 Pairplot — Multivariate View

```
In [26]:
sns.pairplot(data, hue='Product',
vars=['Age', 'Income', 'Miles', 'Fitness'])
plt.show()
```

7.3 Key Correlation Values

Feature Pair	r Value	Strength	Business Insight
Income — Miles	+0.46	Moderate	Higher earners plan more miles; strong KP781 signal
Usage — Miles	+0.41	Moderate	Frequent users plan higher weekly mileage
Usage — Income	+0.38	Moderate	Higher-income customers commit to more sessions

Feature Pair	r Value	Strength	Business Insight
Fitness — Miles	+0.31	Moderate	Fitter customers set more ambitious mileage goals
Age — Miles	+0.12	Weak	Age alone is a poor predictor of usage intensity

■ **Insight: Fitness, Usage, and Miles all correlate above 0.28 with each other — confirming these three variables together form a reliable 3-question POS recommendation tool.**

■ *Note: KP281 and KP481 overlap significantly in the correlation space. A price-sensitivity question is the most reliable way to route customers between these two products.*

KP281: Entry-level buyers with lower income, lower fitness, and lower planned miles. KP481: Mid-range buyers with moderate fitness and usage — overlaps strongly with KP281. KP781: Premium buyers with higher income, higher fitness (4–5), and higher planned miles (100+); this segment is clearly separated.

8. Customer Profiling

KP281 — Entry Level · \$1,500

Age	18–35 years (primary: 18–25)
Gender	Mixed — slight female skew; 55.1% of female buyers choose KP281
Income	\$30K–\$55K — budget-conscious segment
Fitness	1–3 / 5 — beginners or light exercisers
Usage	2–3 sessions/week
Miles/Week	50–120 miles — light to moderate runners
Marital	Slightly more single customers (48% of singles choose KP281)
Education	12–16 years — no strong signal across products
Strategy	Gateway product. High volume, lower margin. Track usage milestones to trigger upgrade path to KP481/KP781.

KP481 — Mid Level · \$1,750

Age	26–50 years (strongest in 36–50 at 44.1%)
Gender	Slight male skew (56% of buyers are male)
Income	\$40K–\$70K — value-oriented mid-segment
Fitness	2–4 / 5 — moderate fitness, improvement goals
Usage	3–4 sessions/week
Miles/Week	80–160 miles — regular runners seeking consistency
Marital	Balanced — partnered customers slightly more represented
Education	13–18 years — similar to KP281; not a key differentiator
Strategy	Overlaps with KP281 on all fitness metrics. Add a budget/price-sensitivity question to route correctly.

KP781 — Premium Level · \$2,500

Age	26–50 years — represented across both working-age groups
Gender	Strongly male — 82% of KP781 buyers are male; $P(KP781 Male)=31.4\%$
Income	\$55K–\$104K — bimodal: \$55K–65K AND \$85K+
Fitness	4–5 / 5 — serious, dedicated fitness users
Usage	4–6 sessions/week — high commitment
Miles/Week	130–210+ miles — competitive or performance runners

Marital	More partnered; $P(KP781 Partnered)=24.8\%$ vs 18.7% for singles
Education	Marginally higher than KP281/KP481 buyers
Strategy	Easiest segment to identify. Income + Miles + Fitness are the top 3 predictors. Two-track messaging by income level.

9. Business Recommendations

1

Target KP781 at High-Income Males (30–45)

$P(KP781|Male)=31.4\%$ vs $P(KP781|Female)=10.3\%$. Focus premium marketing on male professionals aged 30–45 earning >\$60K via LinkedIn, sports apps, and fitness communities. Two messaging tracks: "Performance Upgrade" for \$55K–65K earners and "Elite Fitness" for \$85K+ earners.

2

Grow the Female KP781 Segment — Biggest Untapped Opportunity

Only 1 in 10 female customers buys KP781 — a 3x gap vs males. Female-focused campaigns emphasising performance tracking and health metrics can unlock a significantly under-served premium segment.

3

Use KP281 as a Gateway Product with Upgrade Paths

KP281 holds 44.4% market share. Track usage milestones and trigger personalised KP481/KP781 recommendations when weekly usage consistently exceeds 4 sessions or 150 miles. Turn budget buyers into lifetime AeroFit customers.

4

Build a 3-Question Point-of-Sale Recommendation Tool

Fitness Rating, Weekly Usage, and Planned Miles all correlate above 0.28 and strongly separate KP781. A decision tool asking these 3 inputs + income bracket can instantly surface the right product and reduce purchase friction.

5

Target Partnered Customers for Premium Upgrade

$P(KP781|Partnered)=24.8\%$ vs $P(KP781|Single)=18.7\%$. Home gym and family fitness promotions can convert this segment. Partnered customers likely have dual incomes and longer-term fitness commitment.

6

Add a Budget Question to Distinguish KP281 from KP481

Correlation analysis shows KP281 and KP481 buyers are nearly identical in Fitness, Usage, and Miles. A single price-sensitivity question at point-of-sale is needed to route customers correctly between these two products.

10. Conclusion

Key Finding	Summary
Primary Segment Driver	Income is the strongest predictor — ~48% of >\$60K earners choose KP781
Gender Gap	Males are 3x more likely to buy KP781; female premium segment is underdeveloped
KP281/KP481 Overlap	Needs a price-sensitivity question to disambiguate at point-of-sale
KP781 Clarity	Easiest to identify: High Income + High Fitness + High Miles
Age Signal	Age is a weak predictor; income and fitness are far more discriminating
Education	Not a meaningful differentiator across product segments in this dataset
Actionable Next Step	3-question POS tool (Fitness + Miles + Income bracket) for instant recommendation

The data clearly demonstrates that AeroFit's treadmill market is segmented primarily by **income, gender, fitness level, and planned usage habits** — not age alone. The KP781 segment is the easiest to identify and should be the focus of targeted premium marketing, especially toward the underserved female audience.

AeroFit should invest in a **data-driven recommendation system** at point-of-sale capturing 3–4 key inputs (fitness rating, income range, planned usage, gender) to instantly surface the most appropriate product. KP281 should be positioned as a gateway to the AeroFit ecosystem — turning budget buyers into long-term premium customers through structured upgrade paths.